

Sri Lanka Institute of Information Technology



Web-based Salon Management System

Project ID: RE_G19

Assignment - 01

Submitted by:

Student ID	Student Name
IT24103648	Ranasinghe R A D S
IT24103631	Heiyanthuduwa D D M P
IT24102614	Kahadawala D.C
IT24103389	Perera H K B
IT24103331	Saraniya V.K

1. Introduction about the Project

1.1 Project Domain

This project is about developing a web-based saloon management system to “Harshi Ladies Salon” a local beauty and wellness center offering hair, skin, and cosmetic services, who currently find it difficult to continue with the traditional way of service provision, which makes it challenging for both staff and clients to manage appointments and services efficiently. Salon provides customers with different types of services and introduces packages. Also, appointment bookings are handled manually by customers giving a phone call or making a visit to the salon and all service records, payments, and customer details are maintained manually. The salon provides a variety of individual services as well as service packages, but the existing manual approach leads to inefficiencies, errors, and difficulty in tracking customer preferences and service history. The proposed system falls within the domain of Salon Management Software, which aims to automate and streamline operations such as appointment booking, payment processing, and customer relationship management. The project will provide a digital platform where the salon staff can efficiently manage all operations, reduce errors, and enhance the customer experience.

1.2 Existing Problems

- Difficulty in Customer Record Keeping
 - Maintaining customer history, preferences, and service records in notebooks cause duplicates and redundancies. Retrieving past records for follow-ups or personalized services consumes unnecessary time.
- Operational Inefficiency
 - Staff members spend significant time on administrative tasks rather than focusing on customer service and also face difficulties managing their daily schedules, leading to lower overall productivity.
- Inefficient Appointment Management
 - Scheduling appointments manually leads to double bookings or missed slots. There is no automated reminder system, which results in client no-shows or confusion.
- Inefficient Payment Management
 - Payments and receipts are recorded manually, increasing the risk of mistakes and making it difficult to track financial transactions accurately and reduce the ability to calculate correct profit amounts.

- Inefficient service selection/recommendation
 - A considerable amount of time is wasted for both staff and customers when selecting services such as hairstyles, nail arts, and make-up looks, because there is no proper system in place to pre-select or organize these options in advance.

2. Requirements Elicitation Techniques Used

Requirements elicitation techniques are a crucial part of our project. Requirement elicitation helps to identify and understand what functionalities and features are needed in the system we are building, ensuring that the final product will meet the stakeholders' needs.

To apply these techniques to our projects, first we identified the stakeholders of this business and categorize them into primary, secondary and tertiary stakeholders then conducted **questionnaires**, **interviews** and **observations** to gather the requirements.

1. Why we categorized the stakeholders?

Stakeholders are any person or organization who is affected by the system in some way and who has a legitimate interest. Based on their needs the systems features and functionalities get developed. And helps with understanding which users had the most influence on the system requirements.

2. How the requirement elicitation techniques were applied?

Interview - Interviews are a requirement elicitation technique used to gather detailed information directly from key stakeholders.

We conducted a well-planned interview with the business owner and identified the problems they face with the current business operations. Through this interview, we were able to discover the owner's expectations for the system and the features they require.

Questionnaire - Questionnaires are useful for collecting information from multiple stakeholders efficiently.

For the other stakeholders, we conducted a questionnaire to gather their input. The questionnaire included questions about their roles and the issues they encounter with the current approach. This process helped us identify their precise requirements.

Observation - Observation is a technique used to understand how the current system operates in a real environment.

We also conducted an observation session to examine the business environment and identify other factors to consider when developing the system.

3. How the information helped define requirements?

The requirement elicitation techniques we applied helped us identify key functionalities that the system must provide, such as appointment management, service management, and user management, payment management and service selection.

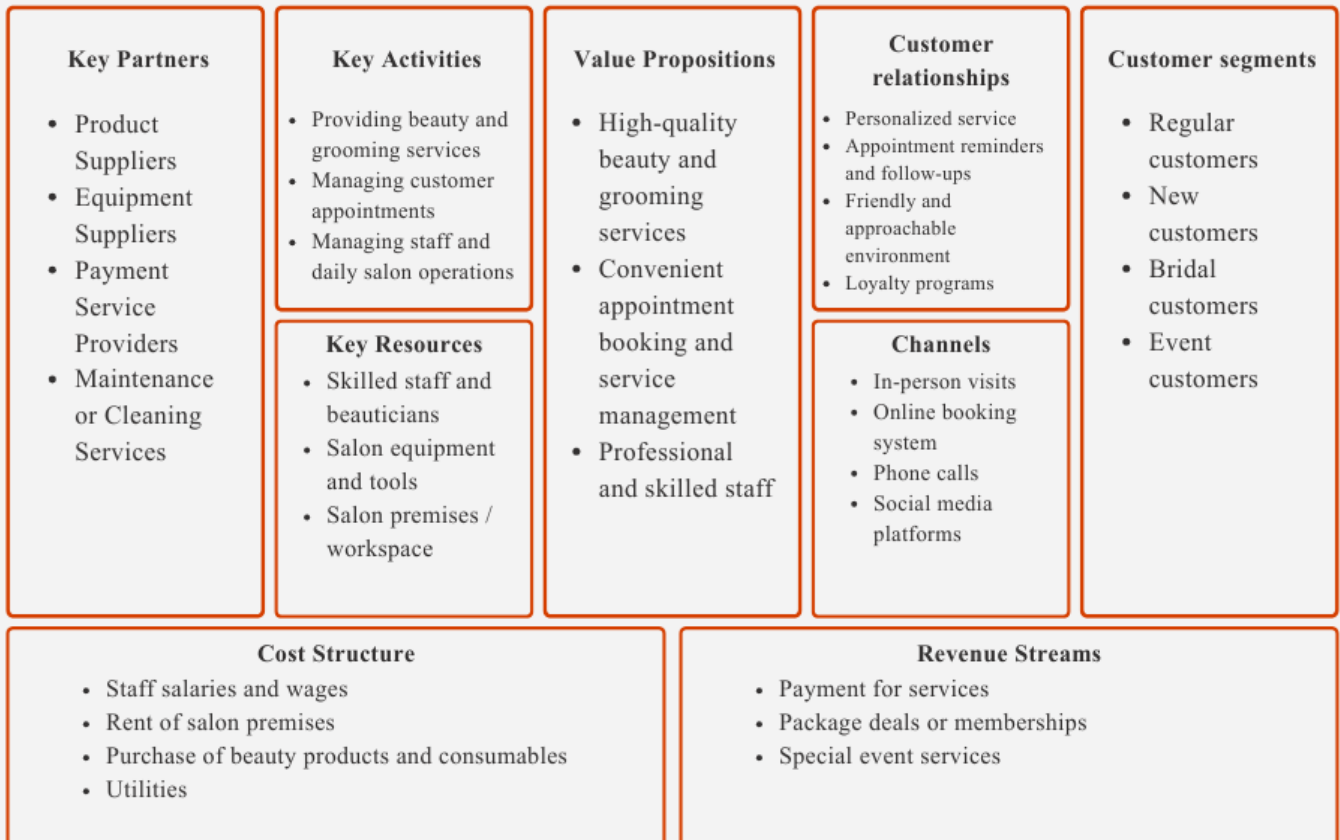
4. Challenges and insights from the techniques

The challenge was that some of the stakeholders were not clear about the technical aspects of the information, and hence we had to define the terms in the questions. However, the process gave us an understanding of the needs of the users.

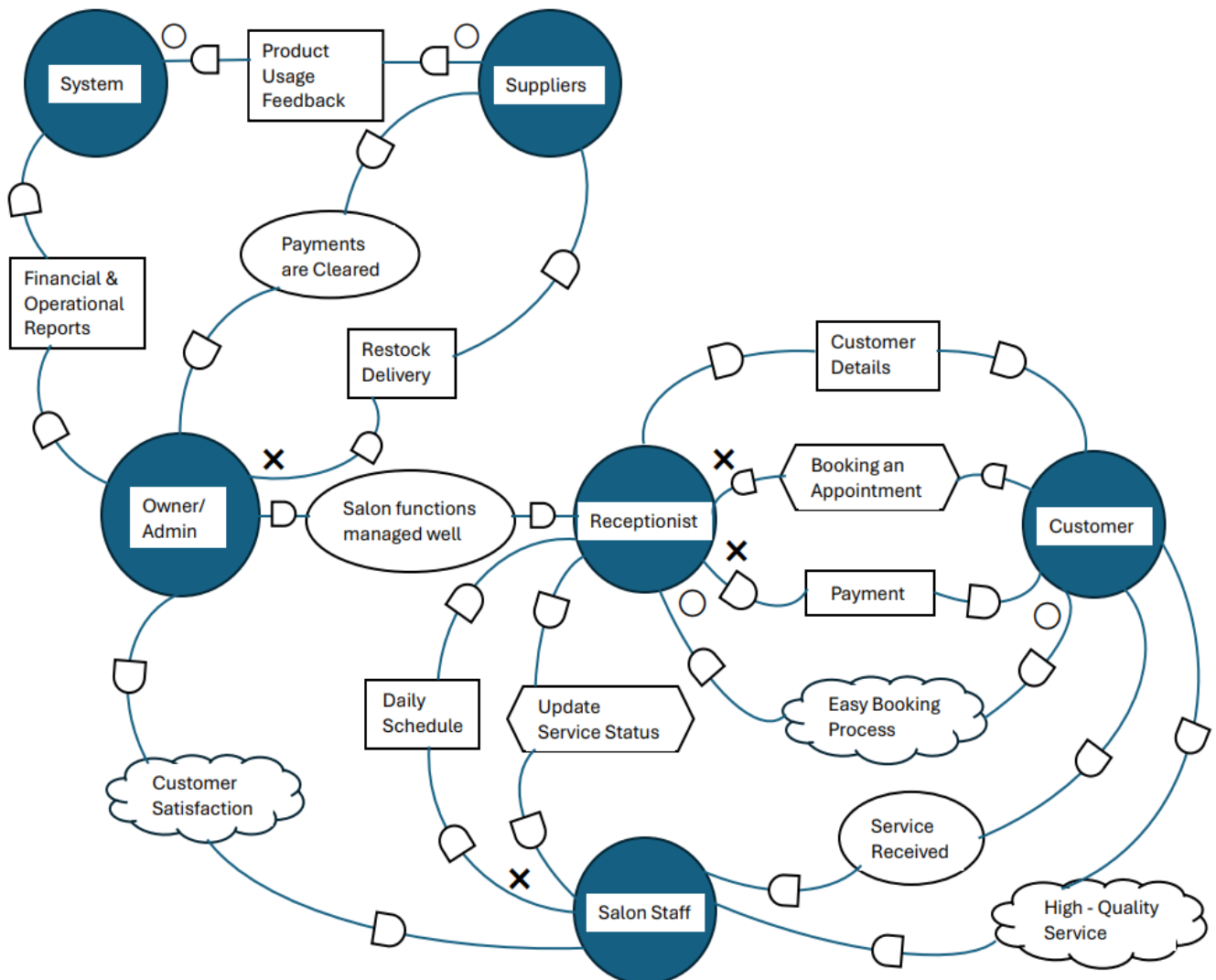
3. Identified Stakeholders

Primary Stakeholders	Secondary Stakeholders	Tertiary Stakeholders
Owner	Product Suppliers	Government/Regulatory Authorities
Staff Members	Equipment Suppliers	Financial Institutions
Receptionist		Local Community
Customer		

4. Business Model Canvas



5. Strategic Dependency model



6. Functional and Nonfunctional requirements

6.1 Functional Requirements

1.User Management

Field	Description
Function name	User Registration
Input	Name, email address, phone number, password
Processing	Validate entered details, check if email already exists, create a new user account
Output	Account created confirmation OR validation error message

Field	Description
Function name	User Login
Input	Email address, password
Processing	Verify credentials against stored user data
Output	User dashboard displayed OR login failure message

2. Service Management

Field	Description
Function name	View Salon Services
Input	Service category selection
Processing	Retrieve service information from system records
Output	List of services with description and price

Field	Description
Function name	Create Service Package
Input	Package name, list of services, package price
Processing	Combine selected services and store the package information
Output	Package creation confirmation

Field	Description
Function name	Add Salon Service
Input	Service name, service category, description, price
Processing	Validate service details and store the service information in the system database
Output	Service successfully added confirmation

Field	Description
Function name	Update Salon Service
Input	Service ID, updated service details
Processing	Retrieve existing service record and update the selected fields
Output	Updated service details confirmation

3.Booking and Appointment Management

Field	Description
Function name	Book Appointment
Input	Customer ID, selected service or package, preferred date, preferred time
Processing	Check staff availability, verify time slot, record appointment details
Output	Appointment confirmation OR unavailable time slot message

Field	Description
Function name	Update or Cancel Appointment
Input	Appointment ID, new date/time or cancel request
Processing	Verify appointment record, update or cancel appointment
Output	Updated appointment confirmation OR cancellation confirmation

Field	Description
Function name	View Staff Availability
Input	Date selection
Processing	Retrieve staff schedules and booked appointments
Output	Available staff and time slots list

5.Feedback and rating system

Field	Description
Function name	Submit Feedback
Input	Customer ID, service ID, staff member ID, rating (1–5), feedback comment
Processing	Validate rating value, store rating and feedback in the system records
Output	Feedback submission confirmation

Field	Description
Function name	Edit Feedback
Input	Feedback ID, updated rating (1–5), updated feedback comment
Processing	Retrieve the existing feedback record, update rating and comment fields
Output	Updated feedback confirmation

Field	Description
Function name	Delete Feedback
Input	Feedback ID
Processing	Locate the feedback record and remove it from the system records
Output	Feedback deletion confirmation

6. Recommendation system

Field	Description
Function name	Delete Feedback
Input	Feedback ID
Processing	Locate the feedback record and remove it from the system records
Output	Feedback deletion confirmation

Field	Description
Function name	View Style Recommendations
Input	User request to view styles
Processing	Retrieve style recommendation records stored in the system
Output	Display list of available hairstyle and beauty style recommendations

Field	Description
Function name	Save Recommended Style
Input	User ID, Style ID
Processing	Store selected style in the user's saved styles list
Output	Saved style confirmation

Field	Description
Function name	Edit Saved Style
Input	User ID, Style ID, updated style preference
Processing	Retrieve saved style record and update the stored preference
Output	Updated style confirmation

Field	Description
Function name	Edit Saved Style
Input	User ID, Style ID, updated style preference
Processing	Retrieve saved style record and update the stored preference
Output	Updated style confirmation

Field	Description
Function name	Delete Saved Style
Input	User ID, Style ID
Processing	Locate saved style record and remove it from the user's saved styles list
Output	Style deletion confirmation

6.2 Non-Functional Requirements

NFR1	Performance
Description	The system shall display requested web pages within 3 seconds for 95% of user requests when accessed by up to 100 concurrent users.

NFR2	Availability
Description	The system shall maintain 99% system availability during operating hours.

NFR3	Security
Description	The system shall store user passwords using encrypted hashing algorithms.

NFR4	Access Control
Description	The system shall restrict system functions based on user roles: customer, staff member, receptionist, and owner.

NFR4	Data Integrity
Description	The system shall prevent double booking by ensuring that only one appointment exists per time slot for a staff member.

6.3 Constraints

1. The system has to be something that we can access through the internet on a computer or laptop.
2. The system must operate on commonly used devices such as laptops or desktop computers.
3. Have to make sure that the system does not let appointment times overlap for the staff members.
4. The system has to keep user data safe and store it in a way so that people's private information is protected.
5. When it comes to payments the system must record them correctly before it confirms any appointments, for the appointment system.

7. Acceptance Criteria

FR1 – Customer Registration

Requirement: The system shall allow new customers to register an account.

Acceptance Criteria

- The system must provide a registration form for customers.
- The customer must enter name, email, phone number, and password.
- The system must validate required fields before submission.
- The system must check that the email is not already registered.
- If registration is successful, the system must display “Registration Successful” and create the customer account.

FR2 – Customer Login

Requirement: The system shall allow registered customers to log in to the system.

Acceptance Criteria

- The customer must enter registered email and password.
- The system must verify the entered credentials.
- If credentials are correct, the system must redirect the user to the dashboard/home page.
- If credentials are incorrect, the system must display an error message.
- The system must allow the user to retry login.

FR3 – Book Salon Appointment

Requirement: The system shall allow customers to book appointments for salon services.

Acceptance Criteria

- The customer must be able to select a service from the service list.
- The customer must choose date and time for the appointment.
- The system must display available time slots only.
- The system must prevent booking conflicts (same time slot).
- After successful booking, the system must display “Appointment Confirmed”.

FR4 – Online Payment for Services

Requirement: The system shall allow customers to make payments for salon services.

Acceptance Criteria

- The system must allow customers to select a payment method.
- The system must verify payment details before processing.
- The system must display payment success or failure message.
- After successful payment, the system must update appointment status to “Paid”.
- The system must generate a payment confirmation or receipt.

FR5 – Admin Manage Salon Services

Requirement: The admin shall be able to manage salon services in the system.

Acceptance Criteria

- The admin must be able to add new services to the system.
- The admin must be able to update service details.
- The admin must be able to delete services.
- The system must display the updated service list after changes.
- Only authorized admin users must be allowed to access service management.

8. UML Use Case diagram and Scenarios

8.1 Use Case Diagram

Web-based Salon Management System



8.2 Epic Wise Use Case Scenarios

1. Customer and Admin Management (User Management)

Field	Description
Number	01
Name	Customer Login
Summary	User's attempt to access system features by login
Priority	High
Preconditions	User having registered to the system User has a valid username and password
Postconditions	Get access to the system features according to the relevant role Customer dashboard is displayed
Primary Actor(s)	Customer, Staff, Receptionist, Admin
Secondary Actor(s)	-
Trigger	User enter username and password

Main Scenario	Step	Action
	1	System displays the login form to enter details.
	2	User enter username and password and enter.
	3	Customer clicks the Login button.
	4	System validates the entered credentials.
	5	System authenticates the customer.
	6	System redirects the customer to the dashboard.

Step	Branching Action
2a	System detects empty fields and asks customer to enter required details
3a	Customer cancels login process and system exit
4a	System notifies login details are invalid and asks customer to try again
5a	System cannot connect to database and shows system error message

2. Package Management

Field	Details
Use Case ID	UC2
Use Case Name	View Service Packages
Actor(s)	Receptionist
Preconditions	Receptionist is registered and logged in with receptionist privileges; System database is operational
Trigger	Receptionist clicks on “View Packages”
Main Flow	1. Receptionist navigates to the Package Management page. 2. Receptionist selects View Packages option. 3. System retrieves the list of available service packages from the database. 4. System displays package details (Package Name, Included Services, Price, Duration). 5. Receptionist reviews the package details to inform customers about available packages.
Alternative Flows	Receptionist searches for a specific package using the search option or filters packages by category (e.g., Bridal Package, Hair Care Package, Facial Package).
Exception Flows	3a. If the system fails to retrieve package data from the database, the system displays an error message “Unable to load packages. Please try again later.”
Postconditions	Receptionist successfully views available service packages and provides package information to customers.

3. Appointment and Payment Management

Item	Details
Use case ID	UC-APM-01
Use Case Name	Book Appointment
Actor(s)	Customer
Preconditions	Customer is registered and logged into the system
Trigger	Customer clicks “Book Appointment”
Main Flow	1. Customer selects service or package 2. System shows available date and time slots 3. Customer selects preferred date and time 4. System checks availability 5. Customer confirms appointment 6. System saves appointment 7. System displays confirmation message
Alternative Flows	Receptionist books appointment on behalf of customer
Exception Flows	4a. Selected time slot is unavailable 4b. System asks customer to choose another slot
Postconditions	Appointment is saved and confirmation is displayed

4. Feedback and Rating System

Field	Example / Details
Use Case ID	UC-05
Use Case Name	Submit Feedback and Rating
Actor(s)	User (Customer/Client)
Preconditions	User is registered and logged in; Service/product has been completed or used
Trigger	User clicks “Give Feedback & Rating”
Main Flow	1. User navigates to Feedback & Rating 2. User enters feedback in text box 3. User selects a rating (1–5 stars) 4. User clicks “Submit” 5. System validates input 6. System saves feedback and rating 7. System displays confirmation message
Alternative Flows	User submits feedback via mobile app instead of website
Exception Flows	3a. Feedback text is empty → system prompts user to enter feedback 3b. Rating not selected → system prompts user to select rating
Postconditions	Feedback and rating are saved; other users/admins can view feedback and rating

5. Style Recommendation System Management

Field	Example / Details
Use Case ID	UC-06
Use Case Name	Manage Style Recommendations
Actor(s)	User (Client/Customer)
Preconditions	User is registered and logged in; System has pre-loaded style recommendations
Trigger	User clicks “View Style Recommendations”
Main Flow	1. User navigates to Style Recommendations section 2. User fills the form in that page 3. System displays recommended styles 4. User views details of a style 5. User saves style to personal list 6. User edits saved style preferences 7. User deletes saved style 8. System confirms each action
Alternative Flows	User saves a style using mobile app instead of website; User edits style notes instead of color
Exception Flows	5a. User tries to save a style already saved → system shows “Style already saved” 6a. User enters invalid data while editing → system shows error message
Postconditions	User’s style preferences are updated and accessible anytime

6. Service Management

Field	Details
Use Case ID	UC1
Use Case Name	Create New Style
Actor(s)	Admin
Preconditions	Admin is registered and logged in with Admin privileges; System database is operational
Trigger	Admin clicks on “Add New Style”
Main Flow	1. Admin navigates to the Salon Service page 2. Admin clicks the "Add New Style" button 3. System displays the style creation form 4. Admin enters style details (e.g., Style Name) 5. Admin clicks “Save” 6. System validates the input data 7. System saves the new style to the database 8. System displays a confirmation message
Alternative Flows	Admin creates a new style by duplicating an existing style and modifying the details before saving.
Exception Flows	4a. If mandatory fields (e.g., Style Name or Price) are empty, system prompts to complete required field. 4b. If the input price is a minus value, the system will ask the admin to input a positive value for the price 6a. If a style with the exact same name already exists in the system, system displays an error message "Style name already exists."
Postconditions	The new style is saved in the database and becomes immediately available to be viewed by customers

9. Product Backlog

1. Admin and Customer Management (User Management)

User Story ID	Remarks	Estimated story points	Estimated hours	Priority (High / Medium / Low)
UM1	Customer account registration	5	8	High
UM2	Customer login	3	5	High
UM6	Owner creates staff & receptionist accounts	5	8	High
UM7	Owner login	3	5	High
UM8	Owner assigns roles to users	5	8	High
UM11	Role-based access control	5	8	High
UM15	Receptionist login	3	5	High
UM20	Staff Member login	3	5	High
UM14	Owner logout	1	2	Medium
UM3	Customer views profile	2	3	Medium
UM4	Customer updates profile	3	4	Medium
UM9	Owner views all users	3	5	Medium
UM10	Owner activates/deactivates users	3	5	Medium

UM12	Owner searches users	3	4	Medium
UM16	Receptionist views profile	2	3	Medium
UM17	Receptionist updates profile	3	4	Medium
UM18	Receptionist views staff roles & availability	3	5	Medium
UM21	Staff views profile	2	3	Medium
UM22	Staff updates profile	3	4	Medium
UM23	Staff changes password	3	4	Medium
UM5	Customer reset password	2	3	Low
UM13	Owner views user activity status	2	3	Low
UM19	Receptionist logout	1	2	Low
UM24	Staff logout	1	2	Low
UM25	Customer views service history	5	8	High

2. Service Management

User Story ID	Remarks	Estimated story points	Estimated hours	Priority (High / Medium / Low)
SM8	Create Service Categories	8	8-9	High
SM1	View Service Categories	3	6-8	High
SM3	Create New Services	8	10-11	High
SM2	View Service Details	5	8-9	High
SM6	View Service lists in each category	5	8-9	Medium
SM4	Update Existing Services	5	9-10	Medium
SM12	Create a Cart	5	8-9	Medium
SM7	Add services to the cart	3	6-7	Medium
SM13	Update the Cart	5	6-7	Medium
SM14	Remove Services from the Cart	3	4-5	Medium
SM11	View Service List in Cart	2	6-7	Low
SM5	Delete Services	3	4-6	Low
SM9	Update Service Categories	3	8-9	Low
SM10	Delete Service Categories	3	4-6	Low

3. Package Management

User Story ID	Remarks	Estimated story points	Estimated hours	Priority (High / Medium / Low)
PM1	Core functionality; allows adding new packages with all details	5	30-32hrs	High
PM2	Needed for customer interaction; simple listing and search	3	16-18hrs	High
PM3	Show price, services, duration; improves UX	2	10-12hrs	High
PM4	Helps staff manage workload and schedule	2	10-12hrs	Medium
PM5	Help increase sales during festivals.	5	28-30hrs	High
PM6	High demand premium service.	5	30hrs	High
PM7	Keeps pricing and services updated.	4	20hrs	Medium
PM8	Helps staff understand requirements	2	10-12hrs	Medium
PM9	Improve marketing and sales	3	15hrs	Medium
PM10	Improve customer experience.	2	10hrs	Medium

4. Appointment and Payment Management

User Story ID	Remarks	Estimated Story Points	Estimated Hours	Priority
APM1	Customers can book appointments by selecting service, date, and time	8	16 hrs	High
APM2	Customer can view upcoming and past appointments	5	10 hrs	High
APM3	Customer can reschedule or update appointment details	8	14 hrs	High
APM4	Customer can cancel appointments	5	8 hrs	High
APM5	Owner can view and manage all appointments	8	15 hrs	High
APM6	Customer can make online payment for appointment	13	22 hrs	High
APM7	Customer can view payment history	5	9 hrs	Medium
APM8	Reception can update payment status (Paid / Pending / Refunded)	5	10 hrs	Medium

5.Feedback and Rating System

User Story ID	Remarks	Estimated story points	Estimated hours	Priority (High / Medium / Low)
FM1	Create detailed comments	3	5	High
FM2	Select a specific staff member	1	2	Low
FM3	Select specific service	1	2	Low
FM4	Edit submitted feed back	2	4	Medium
FM5	Delete inappropriate or fake reviews	8	15	High
FM6	Admin Views ratings	8	15	High

FM7	Staff members View feedback according to service they provided	5	10	High
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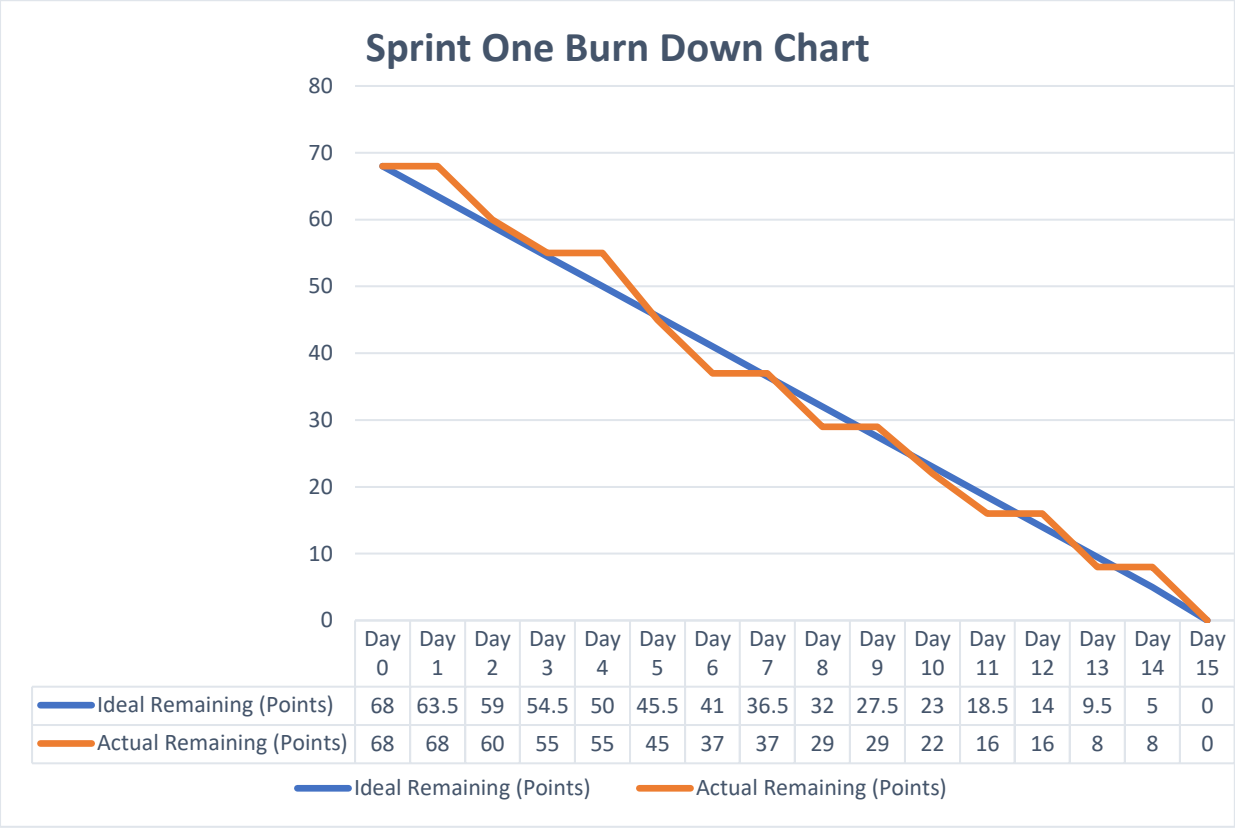
6. Style Recommendation System

User Story ID	Remarks	Estimated story points	Estimated hours	Priority (High / Medium / Low)
S01	Customer identifies styles matching face features and skin tone	5	8	High
S02	Customer views styles that fit facial features and complexion	5	8	High
S03	Admin adds Style Recommendation System to salon system	8	14	High
S04	Customers use recommendations before appointments (staff benefit)	3	5	Medium
S05	System recommends styles using professional rules	8	12	High

10. Burn-down Chart

Day	Ideal Remaining (Points)	Actual Remaining (Points)	Sprint Activity / Features Completed
Day 0	68	68	Sprint Start: Sprint backlog committed (68 pts).
Day 1	63.5	68	Setting up development environments and database schemas.
Day 2	59	60	Completed User accounts and logins (8 pts burned).
Day 3	54.5	55	Completed Role-Based Access Control (5 pts burned). User Management Epic tasks done.
Day 4	50	55	Working on Service Categories CRUD.
Day 5	45.5	45	Completed Service Categories & Salon Services CRUD (10 pts burned).
Day 6	41	37	Completed Styles CRUD (8 pts burned). Service Management tasks done.
Day 7	36.5	37	Working on Package categories.
Day 8	32	29	Completed Create & Read for Packages (8 pts burned). Package tasks done.
Day 9	27.5	29	Working on core Appointment logic.
Day 10	23	22	Completed Create and View Appointments (7 pts burned).
Day 11	18.5	16	Completed Reschedule and Delete Appointments (6 pts burned). Appointment tasks done.
Day 12	14	16	Working on Feedback UI and backend.
Day 13	9.5	8	Completed Create and View Feedback/Ratings (8 pts burned). Feedback tasks done.
Day 14	5	8	Working on Style Recommendation input forms.

Day 15	0	0	Completed Style Recommendation inputs and views (8 pts burned). Sprint Goal Met!
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11. Requirements Traceability Metrix (RTM)

Stakeholder	Requirement ID	User Story ID	Requirement Description	Development Task	Test Case
Customer	FR01	UM1	Customer account registration	Implement user registration form and database storage	Check that a user can actually create an account if they enter their credentials
Customer	FR02	UM2	Customer login	Implement authentication	Verify login with valid username and password
Owner	FR03	UM6	Owner creates staff accounts	Implement staff account creation functions	Verify owner can add new staff
Owner	FR04	SM3	Manage salon services	Implement service CRUD operations	Verify services can be added/updated/deleted
Customer	FR05	SM7	Add services to cart	Implement cart functionality	Verify services appear in cart
Customer	FR06	APM1	Appointment booking	Implement appointment scheduling system	Verify appointment is created successfully
Receptionist	FR07	APM8	Update payment status	Implement payment status update feature	Verify payment status changes correctly
Customer	FR08	APM6	Online payment	Integrate payment processing	Verify successful payment transaction
Customer	FR09	FM1	Submit service feedback	Implement feedback submission form	Verify feedback is saved correctly

Owner	FR10	FM5	Moderate feedback	Implement review management system	Verify admin can delete inappropriate reviews
Customer	FR11	SRM1	Style recommendation	Implement rule-based recommendation feature	Verify system suggests styles based on inputs

12. References List

1. SLIIT IE2021 – Requirement Engineering – Lab sheet 01 document
2. SLIIT IE2091 – Sprint 0 document
3. Draw.io modelling tool

13. Group members contribution table

Student ID	Student Name	Contribution
IT24103648	Ranasinghe R.A.D.S	UML Use Case diagram, Use Case scenarios
IT24103631	Heiyanthuduwa D.D.M.P	Strategic Dependency model, Burn down chart, Use Case scenario
IT24102614	Kahadawala D.C	RTM, Business Model Canvas, List of stakeholders, Introduction, Description, Use Case scenario
IT24103389	Perera H.K.B	Functional and Non-functional requirements, constraints, Use Case scenario
IT24103331	Saraniya V.K	Acceptance Criteria, Use Case scenario